



VP AIR TEST NEET (2025-26)

PART TEST - 08

DROPPER

DURATION: 180 Minutes

Date: 04-01-2026

M. MARKS: 720

ANSWER KEY

1. (1)	46. (2)	91. (2)	136. (4)
2. (2)	47. (1)	92. (3)	137. (1)
3. (1)	48. (3)	93. (1)	138. (2)
4. (2)	49. (4)	94. (2)	139. (4)
5. (1)	50. (2)	95. (1)	140. (1)
6. (1)	51. (3)	96. (3)	141. (3)
7. (2)	52. (2)	97. (2)	142. (1)
8. (2)	53. (1)	98. (3)	143. (4)
9. (2)	54. (4)	99. (1)	144. (1)
10. (3)	55. (2)	100. (1)	145. (1)
11. (1)	56. (3)	101. (2)	146. (3)
12. (1)	57. (3)	102. (2)	147. (1)
13. (1)	58. (1)	103. (3)	148. (1)
14. (3)	59. (3)	104. (1)	149. (4)
15. (1)	60. (4)	105. (3)	150. (4)
16. (1)	61. (4)	106. (3)	151. (2)
17. (1)	62. (2)	107. (3)	152. (4)
18. (1)	63. (1)	108. (3)	153. (4)
19. (2)	64. (3)	109. (3)	154. (2)
20. (1)	65. (2)	110. (2)	155. (1)
21. (2)	66. (4)	111. (4)	156. (4)
22. (1)	67. (2)	112. (1)	157. (1)
23. (1)	68. (4)	113. (2)	158. (2)
24. (2)	69. (2)	114. (2)	159. (3)
25. (1)	70. (1)	115. (4)	160. (3)
26. (2)	71. (1)	116. (1)	161. (1)
27. (3)	72. (1)	117. (1)	162. (4)
28. (1)	73. (1)	118. (2)	163. (1)
29. (3)	74. (2)	119. (3)	164. (2)
30. (2)	75. (4)	120. (4)	165. (3)
31. (1)	76. (4)	121. (3)	166. (4)
32. (4)	77. (2)	122. (3)	167. (3)
33. (1)	78. (3)	123. (4)	168. (2)
34. (2)	79. (2)	124. (3)	169. (2)
35. (1)	80. (3)	125. (4)	170. (4)
36. (3)	81. (4)	126. (2)	171. (2)
37. (2)	82. (1)	127. (1)	172. (4)
38. (2)	83. (1)	128. (2)	173. (4)
39. (3)	84. (1)	129. (1)	174. (2)
40. (3)	85. (3)	130. (2)	175. (3)
41. (3)	86. (1)	131. (4)	176. (3)
42. (4)	87. (4)	132. (3)	177. (1)
43. (3)	88. (1)	133. (1)	178. (2)
44. (1)	89. (1)	134. (2)	179. (1)
45. (2)	90. (1)	135. (2)	180. (3)

HINTS & SOLUTIONS

1. (1)

$$\text{Pitch } P = v \cos \theta \cdot T = v \cos 30^\circ \cdot \frac{2\pi m}{qB}.$$

$$\text{Radius } R = \frac{mv \sin 30^\circ}{qB}.$$

$$\text{Ratio } R/P = \frac{\sin 30^\circ}{2\pi \cos 30^\circ} = \frac{\tan 30^\circ}{2\pi} = \frac{1}{2\pi\sqrt{3}}.$$

$$\text{Thus } R = \frac{P}{2\pi\sqrt{3}}.$$

(NCERT Ed. 2025-26, 12th, Page No. 112)

2. (2)

Magnetic moment is a vector.

$$\vec{M}_{\text{net}} = \vec{M}_1 + \vec{M}_2.$$

Since they are perpendicular,

$$M_{\text{net}} = \sqrt{M^2 + M^2} = \sqrt{2}M.$$

(NCERT Ed. 2025-26, 12th, Page No. 137)

3. (1)

$$r = \frac{\sqrt{2mK}}{qB}$$

$$r_p \propto \frac{\sqrt{1}}{1} \text{ and } r_\alpha \propto \frac{\sqrt{4}}{2} = \frac{2}{2} = 1$$

Ratio is 1:1.

(NCERT Ed. 2025-26, 12th, Page No. 112)

4. (2)

For any uniformly charged rotating body,

$$\frac{M}{L} = \frac{Q}{2m}$$

$$M = iA = (Q\omega / 2\pi)(\pi R^2) \text{ and } L = I\omega = (mR^2)\omega.$$

(NCERT Ed. 2025-26, 12th, Page No. 129)

5. (1)

Square:

$$B = 4 \times \frac{\mu_0 I}{4\pi(L/2)} (\sin 45^\circ + \sin 45^\circ) = \frac{2\sqrt{2}\mu_0 I}{\pi L}.$$

$$\text{Circle: } 2\pi R = 4L \Rightarrow R = 2L/\pi.$$

$$B' = \frac{\mu_0 I}{2R} = \frac{\mu_0 I \pi}{4L}$$

$$B/B' = \frac{2\sqrt{2}\mu_0 I}{\pi L} \times \frac{4L}{\mu_0 I \pi} = \frac{8\sqrt{2}}{\pi^2}.$$

(NCERT Ed. 2025-26, 12th, Page No. 116)

6. (1)

$$\text{Curie's Law: } \chi \propto \frac{1}{T}.$$

$$\chi_1 T_1 = \chi_2 T_2 \Rightarrow 0.001 \times 300 = \chi_2 \times 600.$$

$$\chi_2 = 0.0005$$

(NCERT Ed. 2025-26, 12th, Page No. 148)

7. (2)

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{encl}}$$

$$I_{\text{encl}} = \int_0^{R/2} (kr^2) 2\pi r dr$$

$$= 2\pi k \left[\frac{r^4}{4} \right]_0^{R/2} = \frac{\pi k R^4}{32}$$

$$B \left(2\pi \cdot \frac{R}{2} \right) = \mu_0 \frac{\pi k R^4}{32} \Rightarrow B = \frac{\mu_0 k R^3}{32}$$

(NCERT Ed. 2025-26, 12th, Page No. 118)

8. (2)

$$I = ev / 2\pi R$$

$$B = \frac{\mu_0 I}{2R} = \frac{\mu_0 ev}{4\pi R^2} = 10^{-7} \times (1.6 \times 10^{-19} \times 6 \times 10^6) / (0.01)^2$$

$$= 9.6 \times 10^{-16} \text{ T}$$

(NCERT Ed. 2025-26, 12th, Page No. 116)

9. (2)

$$\text{Given: } H = 2 \times 10^3 \text{ A/m and } \chi = 599.$$

$$\text{Using the formula } B = \mu_0 H (1 + \chi):$$

$$B = (4\pi \times 10^{-7}) \times (2 \times 10^3) \times (1 + 599)$$

$$B = (4\pi \times 10^{-7}) \times (2 \times 10^3) \times (600)$$

$$B = 4\pi \times 10^{-7} \times 1.2 \times 10^6 \text{ T} = 4.8\pi \times 10^{-1} = 0.48\pi \text{ T}$$

(NCERT Ed. 2025-26, 12th, Page No. 146)

10. (3)

For wire segment QR, $\vec{B}$ and $\vec{\ell}$ are both parallel,
 $\therefore F_{QR} = 0$ For wire segment PQ, $\vec{B}$ and $\vec{\ell}$ are perpendicular,
 $\therefore F_{PQ} = \sqrt{2} I l B$ (NCERT Ed. 2025-26, 12th, Page No. 110)

11. (1)

$$\frac{\mu_0 I}{2\pi x} = \frac{\mu_0 (2I)}{2\pi(d-x)}$$

$$\Rightarrow \frac{1}{x} = 2/(d-x) \Rightarrow d-x = 2x \Rightarrow x = \frac{d}{3}$$

(NCERT Ed. 2025-26, 12th, Page No. 118)

12. (1)

$\mu_r < 1$ for diamagnetic materials.

(NCERT Ed. 2025-26, 12th, Page No. 147)

13. (1)

In a radial magnetic field, $\tau = NIAB = k\phi$.

Thus $I = (k / NAB)\phi$, so $I \propto \phi$.

(NCERT Ed. 2025-26, 12th, Page No. 130)

14. (3)

Gauss's Law for Magnetism ($\oint \vec{B} \cdot d\vec{A} = 0$) implies no monopoles and closed loops.

(NCERT Ed. 2025-26, 12th, Page No. 137)

15. (1)

$$B = \frac{\mu_0 I}{4R} = \frac{10^{-7} \times \pi \times 10}{0.02}$$

$$= \frac{3.14 \times 10^{-6}}{0.02} = 1.57 \times 10^{-4} \text{ T}$$

(NCERT Ed. 2025-26, 12th, Page No. 117)

16. (1)

$$B_{axis} = B_0 \frac{R^3}{(R^2 + x^2)^{\frac{3}{2}}} = \frac{B_0}{8}$$

$$(R^2 + x^2)^{\frac{3}{2}} = 8R^3 \Rightarrow R^2 + x^2 = 4R^2$$

$$\Rightarrow x^2 = 3R^2 \Rightarrow x = R\sqrt{3}$$

(NCERT Ed. 2025-26, 12th, Page No. 116)

17. (1)

$$M = \frac{\text{Total Moment}}{\text{Volume}}$$

$$= (10^{10} \times 1.5 \times 9.27 \times 10^{-24}) / (10^{-6})^3$$

$$= 13.9 \times 10^{-14} / 10^{-18} = 1.39 \times 10^5 \text{ A/m.}$$

(NCERT Ed. 2025-26, 12th, Page No. 145)

18. (1)

$$R = \frac{\sqrt{2mqV}}{qB} \propto \sqrt{m}$$

Thus $R^2 \propto m$

$$\frac{m_1}{m_2} = \left(\frac{R_1}{R_2} \right)^2$$

(NCERT Ed. 2025-26, 12th, Page No. 112)

19. (2)

$$\vec{F} = I (\vec{L} \times \vec{B}) = I [L\hat{i} \times B_0(\hat{i} + \hat{j} + \hat{k})]$$

$$= ILB_0(\hat{k} - \hat{j})$$

$$\text{Magnitude} = ILB_0 \sqrt{1^2 + (-1)^2} = \sqrt{2} ILB_0$$

(NCERT Ed. 2025-26, 12th, Page No. 110)

20. (1)

$$B_{end} = \frac{1}{2} \mu_0 n I = \frac{1}{2} (4\pi \times 10^{-7}) \left(\frac{500}{0.5} \right) (5)$$

$$= \pi \times 10^{-7} \times 1000 \times 10 = \pi \times 10^{-3} \text{ T} \approx 3.14 \times 10^{-3} \text{ T}$$

(NCERT Ed. 2025-26, 12th, Page No. 122)

21. (2)

$$\text{Work done} = -MB(\cos \theta - \cos 0^\circ)$$

$$\therefore W_1 = -MB(\cos 90^\circ - \cos 0^\circ) = MB$$

$$\text{And } W_2 = -MB(\cos 60^\circ - \cos 0^\circ) = \frac{MB}{2}$$

(NCERT Ed. 2025-26, 12th, Page No. 139)

22. (1)

$$\tau = MB \sin \theta$$

$$= (IAB) \sin 30^\circ = (2 \times 0.1 \times 0.05 \times 0.1) \times 0.5$$

$$= 0.001 \times 0.5 = 5 \times 10^{-4} \text{ N-m.}$$

(NCERT Ed. 2025-26, 12th, Page No. 125)

23. (1)

$$\tau = MB \sin \theta. \text{ Plane parallel to } B \Rightarrow \theta$$

(between normal and field) = 90° .

$$\tau = (Ia^2) B \sin 90^\circ = Ia^2 B.$$

(NCERT Ed. 2025-26, 12th, Page No. 125)

24. (2)

$$F_{net} = F_1 - F_2 = \frac{\mu_0 I^2 L}{2\pi L} - \frac{\mu_0 I^2 L}{2\pi (2L)}$$

$$= \frac{\mu_0 I^2}{2\pi} (1 - 0.5) = \frac{\mu_0 I^2}{4\pi}$$

(NCERT Ed. 2025-26, 12th, Page No. 123)

25. (1)

Acceleration is not constant because its *direction* changes (it is always $\perp$ to v).

(NCERT Ed. 2025-26, 12th, Page No. 112)

26. (2)

$$B \propto \frac{1}{r}$$

If distance doubles, field becomes half.

$$\frac{0.4}{2} = 0.2G$$

(NCERT Ed. 2025-26, 12th, Page No. 118)

27. (3)

At the midpoint, fields due to both coils are in the same direction.

If one is reversed, they cancel out exactly.

(NCERT Ed. 2025-26, 12th, Page No. 116)

28. (1)

$$M = iA = (ev / 2\pi r)(\pi r^2) = evr / 2$$

$$\text{New } M' = e(2v)(r/2) / 2 = evr / 2 = M$$

(NCERT Ed. 2025-26, 12th, Page No. 129)

29. (3)

At an axial point, the magnetic field $\vec{B}_1$ acts in the direction of the magnetic moment $\vec{M}$ (from South to North pole).

At an equatorial point, the magnetic field $\vec{B}_2$ acts in the opposite direction to the magnetic moment $\vec{M}$ (from North to South pole).

Since they act in opposite directions, the angle between them is 180° .

(NCERT Ed. 2025-26, 12th, Page No. 141)

30. (2)

$$B = \mu_0 H (1 + \chi)$$

$$B = B_0 + B_m$$

$$B_0 = \mu_0 H \text{ (Field due to free currents)}$$

$$B_m = \mu_0 M \text{ (Field due to the material's bound currents/dipoles)}$$

$$B = \mu_0 (H + M)$$

$$\text{Since intensity of magnetization } M = \chi H,$$

$$B = \mu_0 (H + \chi H)$$

$$B = \mu_0 H (1 + \chi)$$

(NCERT Ed. 2025-26, 12th, Page No. 146)

31. (1)

For no deflection,

$$qE = qvB \Rightarrow v = E / B = 20 / 5 = 4.$$

$$\text{Direction: } \vec{E} + \vec{v} \times \vec{B} = 0 \Rightarrow 20\hat{j} + (v_x\hat{i} \times 5\hat{k}) = 0$$

$$\Rightarrow 20\hat{j} - 5v_x\hat{j} = 0 \Rightarrow v_x = 4$$

(NCERT Ed. 2025-26, 12th, Page No. 112)

32. (4)

For long hollow conducting cylinder,

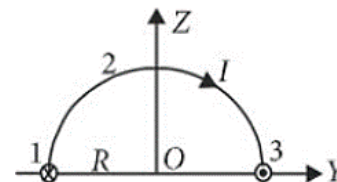
$$B_{\text{inside}} = 0$$

$$B_{\text{outside}} = \frac{\mu_0 i}{2\pi r}$$

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33. (1)

Given situation is shown in the figure.



Parallel wires 1 and 3 are semi-infinite, so magnetic field at O due to them

$$\vec{B}_1 = \vec{B}_3 = -\frac{\mu_0 I}{4\pi R} \hat{k}$$

Magnetic field at O due to semi-circular arc in YZ-plane is given by

$$\vec{B}_2 = -\frac{\mu_0 I}{4R} \hat{i}$$

Net magnetic field at point O is given by

$$\begin{aligned} \vec{B} &= \vec{B}_1 + \vec{B}_2 + \vec{B}_3 \\ &= -\frac{\mu_0 I}{4\pi R} \hat{k} - \frac{\mu_0 I}{4R} \hat{i} - \frac{\mu_0 I}{4\pi R} \hat{k} \\ &= -\frac{\mu_0 I}{4\pi R} (\pi \hat{i} + 2\hat{k}) \end{aligned}$$

(NCERT Ed. 2025-26, 12th, Page No. 117)

34. (2)

At the Curie temperature (T_c), the thermal agitation becomes high enough to disrupt the alignment of magnetic domains in a ferromagnetic substance. Above this temperature, the material loses its spontaneous magnetization and begins to obey Curie's Law, behaving like a paramagnetic substance.

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35. (1)

$$B_{\text{axial}} = \frac{\mu_0}{4\pi} \frac{(2Mr)}{(r^2 - l^2)^2}$$

$$= \frac{10^{-7} \times 2 \times 12 \times 0.2}{[(0.2)^2 - (0.05)^2]^2} = 3.4 \times 10^{-4} \text{ T}$$

(NCERT Ed. 2025-26, 12th, Page No. 140)

36. (3)

$$\tau = MB \sin \theta$$

$$= (4 \times 10^{-6}) \times 2 \times 10^{-5} \times \sin 90$$

$$= 8 \times 10^{-11} \text{ Nm}$$

(NCERT Ed. 2025-26, 12th, Page No. 139)

37. (2)

The component of velocity parallel to the field is $v_{\parallel} = v_0 \hat{j}$. Since the magnetic force $\vec{F} = q(\vec{v} \times \vec{B})$ is zero for this component, $v_{\parallel}$ remains constant throughout the motion.

The component perpendicular to the field is $v_{\perp} = v_0 \hat{i}$. This component will undergo uniform circular motion in the XZ -plane (perpendicular to $\hat{j}$).

$$\omega = \frac{qB_0}{m} = \alpha B_0.$$

$$t = \frac{\pi}{\alpha B_0} = \frac{\pi}{\omega}. \text{ This corresponds exactly to half a}$$

revolution

In circular motion, after half a revolution, the velocity vector in the plane of rotation reverses its direction. Thus, the perpendicular component $v_0 \hat{i}$ becomes $-v_0 \hat{i}$.

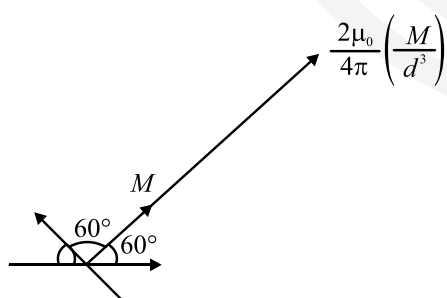
$$\vec{v}_{final} = -v_0 \hat{i} + v_0 \hat{j} = v_0 (-\hat{i} + \hat{j}).$$

Thus, option (2) is correct.

(NCERT Ed. 2025-26, 12th, Page No. 112)

38. (2)

Since two equal vector M are inclined at 120° , their resultant will also be M and along its angular bisector. So point P is on axial line of resultant moment M .



$$B_{net} = \frac{2\mu_0 M}{4\pi d^3}$$

(NCERT Ed. 2025-26, 12th, Page No. 140)

39. (3)

For stable equilibrium, $\theta = 0^\circ$

$$U = -MB$$

$$= -0.32 \times 0.15 = -4.8 \times 10^{-2} \text{ J}$$

(NCERT Ed. 2025-26, 12th, Page No. 139)

40. (3)

$$\text{Lorentz force} = q(\vec{v} \times \vec{B})$$

$$-8\hat{k} = q \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 0 \\ 0 & 2 & 0 \end{vmatrix} \times 10^6$$

$$-8\hat{k} = q\{\hat{i}(0) - \hat{j}(0) + \hat{k}(4-0)\} \times 10^6$$

$$-8\hat{k} = 4q\hat{k} \times 10^6$$

$$\therefore 4q = -8 \times 10^{-6}$$

$$q = -2 \times 10^{-6} \text{ C}$$

$$= -2\mu\text{C}$$

(NCERT Ed. 2025-26, 12th, Page No. 110)

41. (3)

Diamagnetic materials develop a weak induced magnetic moment in a direction opposite to the applied field, leading to feeble repulsion.

Paramagnetic and Ferromagnetic materials are attracted by the field (feeblely and strongly, respectively).

(NCERT Ed. 2025-26, 12th, Page No. 147)

42. (4)

$$T = BIR$$

(NCERT Ed. 2025-26, 12th, Page No. 110)

43. (3)

Magnetic field along the axis of wire length is zero. Point A and C lies on axis of wire.

(NCERT Ed. 2025-26, 12th, Page No. 114)

44. (1)

$$\text{Magnetic field of solenoid } B = \mu_0 nI.$$

$$\text{Magnetic moment of coil } m = NiA.$$

Since axes are perpendicular ($\theta = 90^\circ$), torque

$$\tau = mB \sin 90^\circ = (NiA)(\mu_0 nI) = \mu_0 nINiA.$$

(NCERT Ed. 2025-26, 12th, Page No. 125)

45. (2)

$$\mu_r = 1 + \chi$$

Diamagnetic (1-B): Since χ is small and negative (e.g., -0.00001), μ_r is less than 1.

Paramagnetic (2-C): Since χ is small and positive, μ_r is slightly greater than 1.

Ferromagnetic (3-A): Since χ is very large and positive, μ_r is much greater than 1 (often in the range of 10^3 to 10^5).

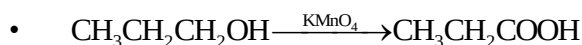
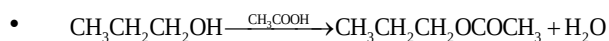
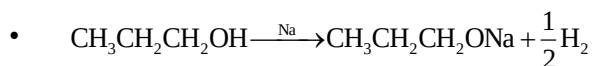
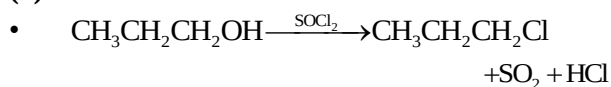
(NCERT Ed. 2025-26, 12th, Page No. 147)

46. (2)

The reactivity order of alcohols towards Lucas reagent: $3^\circ > 2^\circ > 1^\circ$

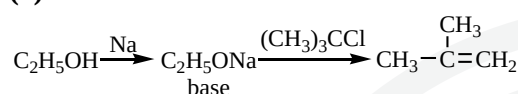
(NCERT Ed. 2025-26 12th Part-II Page No. 208)

47. (1)



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48. (3)

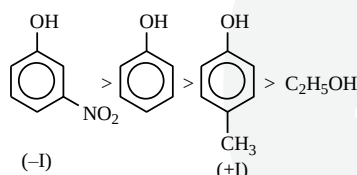


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49. (4)

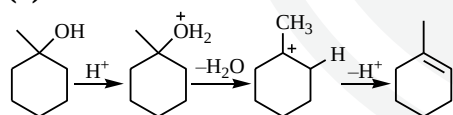
- Lesser the pK_a value, more will be the acidic nature.

- Acidic nature:



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50. (2)



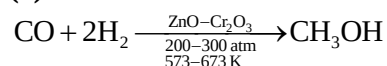
(NCERT Ed. 2025-26 12th Part-II Page No. 209)

51. (3)

HBr give addition reaction to double bond and cleavage of C–O bond in ether take place.

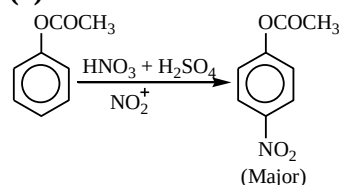
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52. (2)



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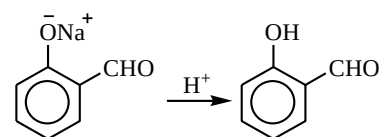
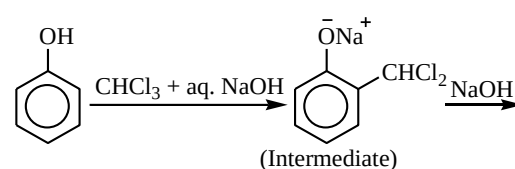
53. (1)



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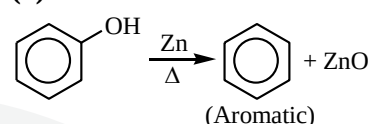
54. (4)

Reimer-Tiemann reaction:



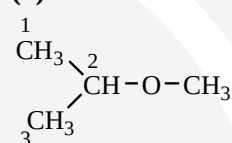
(NCERT Ed. 2025-26 12th Part-II Page No. 213)

55. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 213)

56. (3)



2-Methoxypropane

(NCERT Ed. 2025-26 12th Part-II Page No. 197)

57. (3)

- Phenetole
- Catechol
- Quinol
- Anisole

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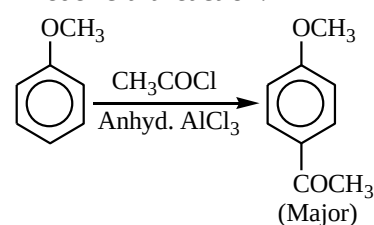
58. (1)

With concentrated nitric acid phenol-2,4-disulphonic acid gives 2,4,6-trinitrophenol

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59. (3)

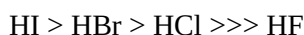
Friedel-Craft reaction:



(NCERT Ed. 2025-26 12th Part-II Page No. 220)

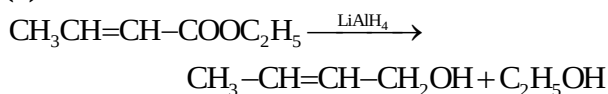
60. (4)

The order of reactivity of hydrogen halides with ethers is:



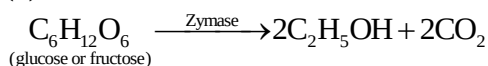
(NCERT Ed. 2025-26 12th Part-II Page No. 217)

61. (4)



(NCERT Ed. 2025-26 12th Part-II Page No. 200)

62. (2)



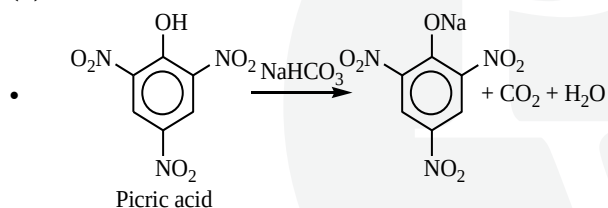
(NCERT Ed. 2025-26 12th Part-II Page No. 214)

63. (1)

- A better reagent for oxidation of primary alcohols to aldehydes in good yield is PCC.
- Secondary alcohols are oxidised to ketones by CrO_3

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64. (3)



- Only those acids give CO_2 on reaction with NaHCO_3 which are more acidic than H_2CO_3 .

(NCERT Ed. 2025-26 12th Part-II Page No. 207)

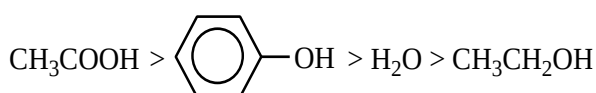
65. (2)

Diazonium salts are hydrolysed to phenol by warming with water.

(NCERT Ed. 2025-26 12th Part-II Page No. 202)

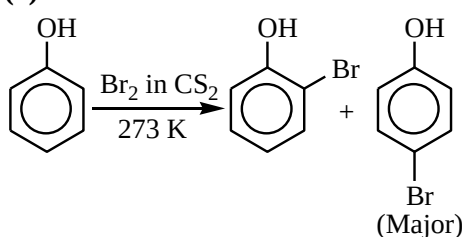
66. (4)

Acidic nature:



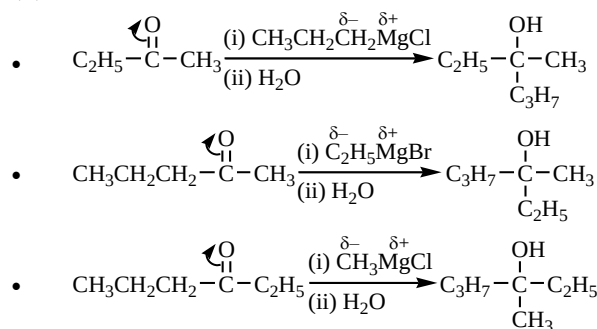
(NCERT Ed. 2025-26 12th Part-II Page No. 205)

67. (2)



(NCERT Ed. 2025-26 12th Part-II Page No. 212)

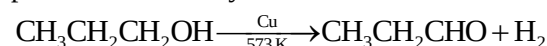
68. (4)



(NCERT Ed. 2025-26 12th Part-II Page No. 201)

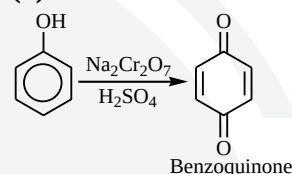
69. (2)

When the vapours of 1° or 2° alcohols are passed over heated copper at 573 K, dehydrogenation takes place and an aldehyde or a ketone is formed.



(NCERT Ed. 2025-26 12th Part-II Page No. 210)

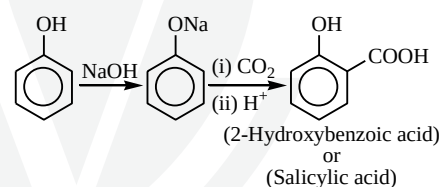
70. (1)



(NCERT Ed. 2025-26 12th Part-II Page No. 213)

71. (1)

Kolbe's reaction:



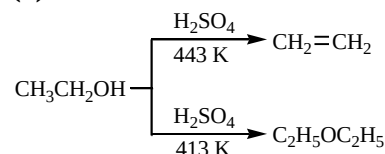
(NCERT Ed. 2025-26 12th Part-II Page No. 212)

72. (1)

Best results in Williamson's synthesis for the preparation of ethers are obtained if the alkyl halide is 1° . In case of 2° and 3° -alkyl halides, elimination competes over substitution.

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73. (1)



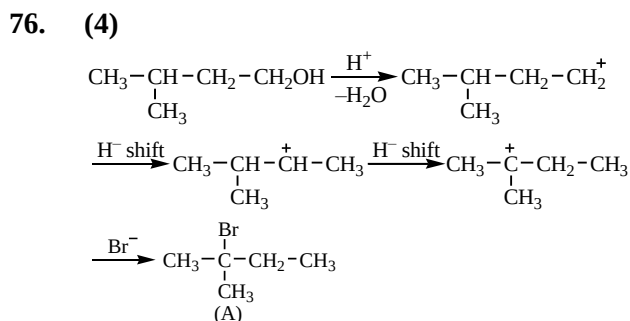
(NCERT Ed. 2025-26 12th Part-II Page No. 215)

74. (2)

The commercial alcohol is made unfit for drinking by mixing in it pyridine and copper sulphate

(NCERT Ed. 2025-26 12th Part-II Page No. 214)

75. (4)
- Phenoxide ion is more reactive than phenol towards electrophilic aromatic substitution.
 - Methanol is also known as wood spirit.
- (NCERT Ed. 2025-26 12th Part-II Page No. 214)

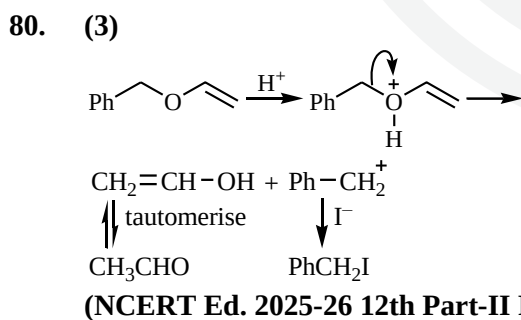


(NCERT Ed. 2025-26 12th Part-II Page No. 208)

77. (2)
- The C–O–C, bond angle in CH₃OCH₃ is slightly greater than the tetrahedral angle due to repulsive interaction between the two methyl groups.
- (NCERT Ed. 2025-26 12th Part-II Page No. 198)

78. (3)
- The relative ease of dehydration of alcohols is: 3° > 2° > 1°
 - The stability order of intermediate carbocation during dehydration of alcohols is: 3° > 2° > 1°
- (NCERT Ed. 2025-26 12th Part-II Page No. 209)

79. (2)
- o-Nitrophenol and p-nitrophenol can be separated by steam distillation.
- (NCERT Ed. 2025-26 12th Part-II Page No. 211)



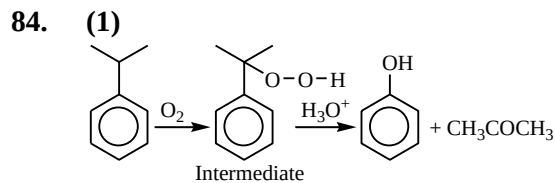
81. (4)
- Phenol is known as carboic acid.
- (NCERT Ed. 2025-26 12th Part-II Page No. 214)

82. (1)
- Acidic nature of alcohols: 1° > 2° > 3°.
- (NCERT Ed. 2025-26 12th Part-II Page No. 205)

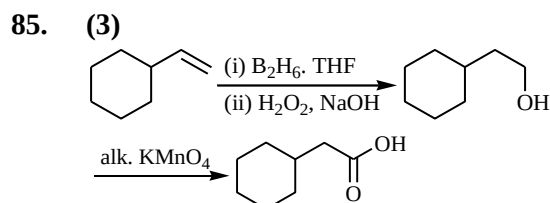
83. (1)
- The solubility of alcohols in water decreases with increase in size of alkyl group.

- Alcohols react both as nucleophiles and electrophile.

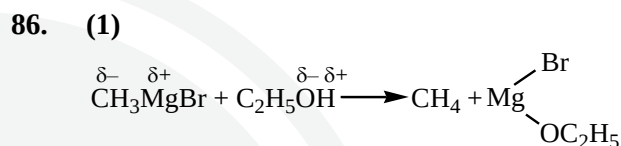
(NCERT Ed. 2025-26 12th Part-II Page No. 204)



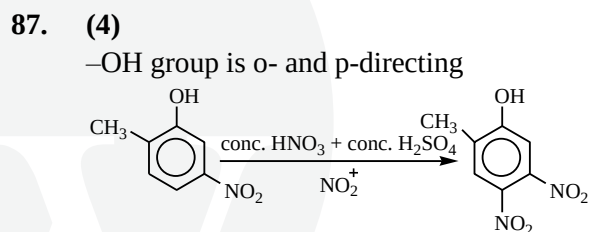
(NCERT Ed. 2025-26 12th Part-II Page No. 202)



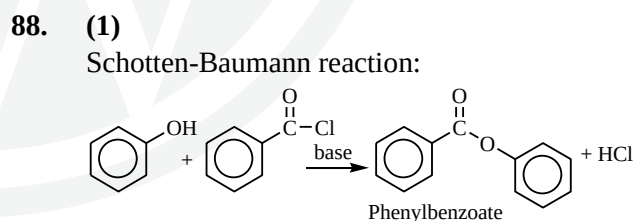
(NCERT Ed. 2025-26 12th Part-II Page No. 200)



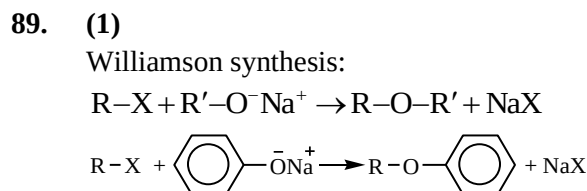
(NCERT Ed. 2025-26 12th Part-II Page No. 205)



(NCERT Ed. 2025-26 12th Part-II Page No. 212)



(NCERT Ed. 2025-26 12th Part-II Page No. 207)



It involve S_N2 attack of an alkoxide ion on 1° alkyl halide

(NCERT Ed. 2025-26 12th Part-II Page No. 215)

90. (1)
- Boiling point of ethanol is more than that of boiling point of methoxy ethane. It is due to presence of the intermolecular hydrogen bonding in ethanol.
- (NCERT Ed. 2025-26 12th Part-II Page No. 204)

91. (2)
Sexual mode of reproduction enables creation of new variants, so that survival advantage is enhanced.
(NCERT Ed. 2025-26, 12th, Page No. 1)
92. (3)
A typical angiosperm anther is bilobed with each lobe having two theca, i.e., they are dithecous. The anther is a four-sided (tetragonal) structure consisting of four microsporangia located at the corners, two in each lobe.
(NCERT Ed. 2025-26, 12th, Page No. 5)
93. (1)
Occasionally, in some seeds such as black pepper and beet, remnants of nucellus are also persistent. This residual, persistent nucellus is the perisperm. The residual, persistent nucellus is the perisperm. The wall of the ovary develops into the wall of fruit called pericarp. Epiblast is the remnant of second cotyledon in grass embryo. The portion of embryonal axis above the level of cotyledons in dicot embryo is the epicotyl.
(NCERT Ed. 2025-26, 12th, Page No. 19, 20)
94. (2)
Panchanan Maheshwari established the Department of Botany, University of Delhi as an important centre of research in embryology and tissue culture. He established the Department of Botany, University of Delhi as an important centre of research in embryology and tissue culture. He also emphasised the need for initiation of work on artificial culture of immature embryos. These days, tissue culture has become a landmark in science. His work on test tube fertilisation and intra-ovarian pollination won worldwide acclaim. He was honoured with fellowship of Royal Society of London (FRS), Indian National Science Academy and several other institutions of excellence.
(NCERT Ed. 2025-26, 12th, Page No. 2)
95. (1)
Cleistogamous flowers do not open at all. In such flowers, the anthers and stigma lie close to each other. Thus, cleistogamous flowers are invariably autogamous as there is no chance of cross-pollen landing on the stigma. Cleistogamous flowers produce assured seed-set even in the absence of pollinators.
(NCERT Ed. 2025-26, 12th, Page No. 12)
96. (3)
The cylindrical portion below the level of cotyledons in a dicot embryo is hypocotyl that terminates at its lower end in the radicle or root tip.
(NCERT Ed. 2025-26, 12th, Page No. 19)
97. (2)
Artificial hybridisation: If the female parent bears bisexual flowers, removal of anthers from the flower bud before the anther dehisces using a pair of forceps is necessary. This step is referred to as emasculation. Emasculated flowers have to be covered with a bag of suitable size, generally made up of butter paper, to prevent contamination of its stigma with unwanted pollen. This process is called bagging. When the stigma of bagged flower attains receptivity, mature pollen grains collected from anthers of the male parent are dusted on the stigma, and the flowers are rebagged, and the fruits allowed to develop.
(NCERT Ed. 2025-26, 12th, Page No. 17)
98. (3)
Non-albuminous seeds have no residual endosperm (e.g., pea, groundnut). Albuminous seeds retain a part of endosperm (e.g., wheat, maize, barley, castor). In a few species such as apple, strawberry, cashew, etc., the thalamus also contributes to fruit formation. Such fruits are called false fruits. The fruits which develop without fertilisation are called parthenocarpic fruits. Banana is one such example.
(NCERT Ed. 2025-26, 12th, Page No. 21)
99. (1)
In angiosperms, endosperm development precedes embryo development. The cells of this tissue are filled with reserve food materials and are used for the nutrition of the developing embryo.
(NCERT Ed. 2025-26, 12th, Page No. 18)
100. (1)
In over 60 per cent of angiosperms, pollen grains are shed at the 2-celled stage.
(NCERT Ed. 2025-26, 12th, Page No. 7)
101. (2)
The ability of the pistil to recognise the pollen followed by its acceptance or rejection is the result of a continuous dialogue between pollen grain and the pistil. This dialogue is mediated by chemical components of the pollen interacting with those of the pistil.
(NCERT Ed. 2025-26, 12th, Page No. 16)

102. (2)
Pollination by water is quite rare in flowering plants and is limited to about 30 genera, mostly monocotyledons. Both wind and water pollinated flowers are not very colourful and do not produce nectar.

(NCERT Ed. 2025-26, 12th, Page No. 13)

103. (3)
Leaf cell is diploid ($2n$) = 20, so the haploid number (n) is 10. The other male gamete moves towards the two polar nuclei located in the central cell and fuses with them to produce a triploid primary endosperm nucleus (PEN). Therefore, $PEN = 3n = 3 \times 10 = 30$.

(NCERT Ed. 2025-26, 12th, Page No. 18)

104. (1)
Although geitonogamy is functionally cross-pollination involving a pollinating agent, genetically it is similar to autogamy since the pollen grains come from the same plant. It involves a pollinating agent.

(NCERT Ed. 2025-26, 12th, Page No. 12)

105. (3)
In some species, the diploid egg cell is formed without reduction division and develops into the embryo without fertilisation. This is called apomixis.

(NCERT Ed. 2025-26, 12th, Page No. 22)

106. (3)
The hard outer layer called the exine is made up of sporopollenin which is one of the most resistant organic material known.

(NCERT Ed. 2025-26, 12th, Page No. 7)

107. (3)
The synergids have special cellular thickenings at the micropylar tip called filiform apparatus, which play an important role in guiding the pollen tubes into the synergid.

(NCERT Ed. 2025-26, 12th, Page No. 11)

108. (3)
Flowering plants have developed many devices to discourage self-pollination and to encourage cross-pollination. In some species, pollen release and stigma receptivity are not synchronised. In some other species, the anther and stigma are placed at different positions. Both these devices prevent autogamy. The third device is self-incompatibility. This is a genetic mechanism and prevents self-pollen from fertilising the ovules. If both male and female flowers are present on the same plant such as castor and maize (monoecious), it prevents autogamy but not geitonogamy.

(NCERT Ed. 2025-26, 12th, Page No. 15)

109. (3)
Four wall layers, the epidermis, endothecium, middle layers and the tapetum surround the microsporangium. The outer three wall layers the epidermis, endothecium and middle layers perform the function of protection and help in dehiscence of anther. The innermost wall layer tapetum nourishes the developing pollen grains. A group of compactly arranged homogenous cells called the sporogenous tissue occupies the centre of each microsporangium. Each one is a potential pollen or microspore mother cell. Pollen grain exine has prominent apertures called germ pores where sporopollenin is absent.

(NCERT Ed. 2025-26, 12th, Page No. 5, 7)

110. (2)
The process of formation of microspores from a pollen mother cell (PMC) through meiosis is called microsporogenesis. Initially, a single diploid ($2n$) pollen mother cell, also known as a microspore mother cell, undergoes meiosis. This single division results in the formation of four haploid (n) microspores, collectively arranged in a tetrad. Following this, each of these four microspores develops into a pollen grain. Within each pollen grain, the nucleus undergoes a mitotic division to form two cells: a vegetative cell and a smaller generative cell. The generative cell then divides mitotically once more to produce two male gametes. Therefore, since one pollen mother cell gives rise to four pollen grains, and each pollen grain contains two male gametes, the total number of male gametes formed is eight (4 pollen grains $\times$ 2 male gametes/pollen grain).

(NCERT Ed. 2025-26, 12th, Page No. 6, 7)

111. (4)
Some examples of water pollinated plants are *Vallisneria*, *Hydrilla* and *Zostera*.

(NCERT Ed. 2025-26, 12th, Page No. 13)

112. (1)
*Orobanch*e and *Striga* are parasitic plants.

(NCERT Ed. 2025-26, 12th, Page No. 22)

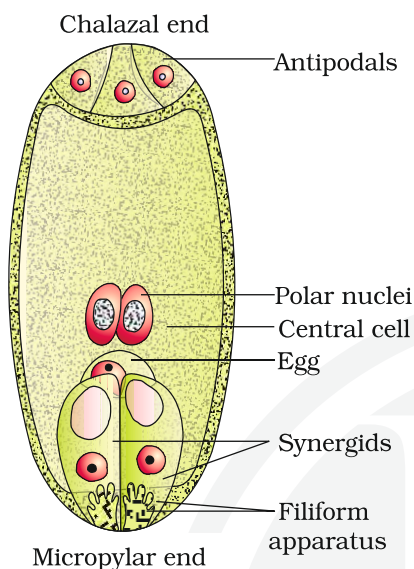
113. (2)
Wind-pollinated flowers often have a single ovule in each ovary and numerous flowers packed into an inflorescence; a familiar example is the corn cob – the ears you see are nothing but the stigma and style which wave in the wind to trap pollen grains.

(NCERT Ed. 2025-26, 12th, Page No. 13)

114. (2)
The hard outer layer of pollen grain called the exine is made up of sporopollenin. Pollen grains are well-preserved as fossils because of the presence of sporopollenin. It is possible to store pollen grains of a large number of species for years in liquid nitrogen (-196°C).

(NCERT Ed. 2025-26, 12th, Page No. 7, 8)

115. (4)



A diagrammatic representation of the mature embryo sac

(NCERT Ed. 2025-26, 12th, Page No. 10)

116. (1)
In some cereals such as rice and wheat, pollen grains lose viability within 30 minutes of their release.

(NCERT Ed. 2025-26, 12th, Page No. 8)

117. (1)
The MMC undergoes meiotic division that results in the production of four megaspores. In a majority of flowering plants, one of the megaspores is functional while the other three degenerate. The nucleus of the functional megaspore divides mitotically to form two nuclei. Two more sequential mitotic nuclear divisions result in the formation of the 4-nucleate and later the 8-nucleate stages. Thus, a typical angiosperm embryo sac, at maturity, though 8-nucleate is 7-celled.

(NCERT Ed. 2025-26, 12th, Page No. 10, 11)

118. (2)
One of the male gametes moves towards the egg cell and fuses with its nucleus thus completing the syngamy. The other male gamete moves towards the two polar nuclei located in the central cell and fuses with them. As this involves the fusion of three haploid nuclei it is termed triple fusion, it is termed triple fusion.

(NCERT Ed. 2025-26, 12th, Page No. 18)

119. (3)
The basal bulged part of the pistil is the ovary.
(NCERT Ed. 2025-26, 12th, Page No. 8)

120. (4)
The body of the ovule fuses with funicle in the region called hilum. A multicarpellary syncarpous gynoecium is present in *Papaver*. The cells of nucellus are diploid.

(NCERT Ed. 2025-26, 12th, Page No. 9)

121. (3)
Production of hybrid seeds is costly and hence the cost of hybrid seeds become too expensive for the farmers. If these hybrids are made into apomicts, there is no segregation of characters in the hybrid progeny. Then the farmers can keep on using the hybrid seeds to raise new crop year after year and he does not have to buy hybrid seeds every year.

(NCERT Ed. 2025-26, 12th, Page No. 23)

122. (3)
Some plants such as *Viola* (common pansy), *Oxalis*, and *Commelina* produce two types of flowers – chasmogamous flowers which are similar to flowers of other species with exposed anthers and stigma, and cleistogamous flowers which do not open at all. In several species such as papaya, male and female flowers are present on different plants, that is each plant is either male or female (dioecy). This condition prevents both autogamy and geitonogamy.

(NCERT Ed. 2025-26, 12th, Page No. 12)

123. (4)
Some examples of water pollinated plants are *Vallisneria* and *Hydrilla* which grow in fresh water and several marine sea-grasses such as *Zostera*. In some species floral rewards are in providing safe places to lay eggs; an example is that of the tallest flower of *Amorphophallus* (the flower itself is about 6 feet in height). A similar relationship exists between a species of moth and the plant *Yucca* where both species – moth and the plant – cannot complete their life cycles without each other.

(NCERT Ed. 2025-26, 12th, Page No. 13, 14)

124. (3)
The flowers pollinated by flies and beetles secrete foul odours to attract these animals.

(NCERT Ed. 2025-26, 12th, Page No. 14)

125. (4)

Wind pollination requires that the pollen grains are light and non-sticky. They often possess well-exposed stamens and large often-feathery stigma. Both wind and water pollinated flowers are not very colourful and do not produce nectar.

(NCERT Ed. 2025-26, 12th, Page No. 13)

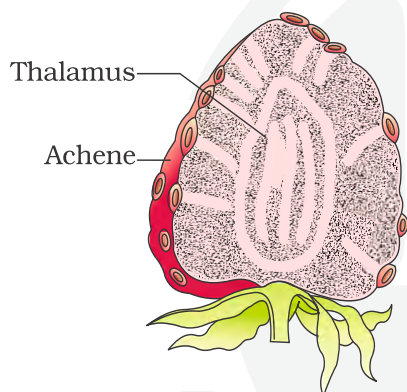
126. (2)

The coconut water from tender coconut that you are familiar with, is nothing but free-nuclear endosperm (made up of thousands of nuclei) and the surrounding white kernel is the cellular endosperm.

(NCERT Ed. 2025-26, 12th, Page No. 19)

127. (1)

In a few species such as apple, strawberry, cashew, etc., the thalamus also contributes to fruit formation. Such fruits are called false fruits.



False fruit of strawberry

(NCERT Ed. 2025-26, 12th, Page No. 20, 21)

128. (2)

The fruits may be fleshy as in guava, orange, mango, etc., or may be dry, as in groundnut, and mustard, etc.

(NCERT Ed. 2025-26, 12th, Page No. 20)

129. (1)

Although seeds, in general are the products of fertilisation, a few flowering plants such as some species of *Asteraceae* and grasses, have evolved a special mechanism, to produce seeds without fertilisation, called apomixis. Thus, apomixis is a form of asexual reproduction that mimics sexual reproduction. There are several ways of development of apomictic seeds. Apomicts have several advantages in horticulture and agriculture.

(NCERT Ed. 2025-26, 12th, Page No. 22, 23)

130. (2)

The oldest viable seed is that of a lupine, *Lupinus arcticus* excavated from Arctic Tundra.

(NCERT Ed. 2025-26, 12th, Page No. 22)

131. (4)

Parthenium or carrot grass that came into India as a contaminant with imported wheat.

(NCERT Ed. 2025-26, 12th, Page No. 7)

132. (3)

When the pollen grain is mature it contains two cells, the vegetative cell and generative cell. The vegetative cell is bigger, has abundant food reserve and a large irregularly shaped nucleus. The generative cell is small and floats in the cytoplasm of the vegetative cell. It is spindle shaped with dense cytoplasm and a nucleus.

(NCERT Ed. 2025-26, 12th, Page No. 7)

133. (1)

Following double fertilisation, events of maturation of ovule(s) into seed(s) and ovary into fruit. The central cell after triple fusion becomes the primary endosperm cell (PEC) and develops into the endosperm while the zygote develops into an embryo.

(NCERT Ed. 2025-26, 12th, Page No. 18, 20)

134. (2)

The pollen grains represent the male gametophytes. Pollen consumption has been claimed to increase the performance of athletes and race horses.

(NCERT Ed. 2025-26, 12th, Page No. 6, 7, 8)

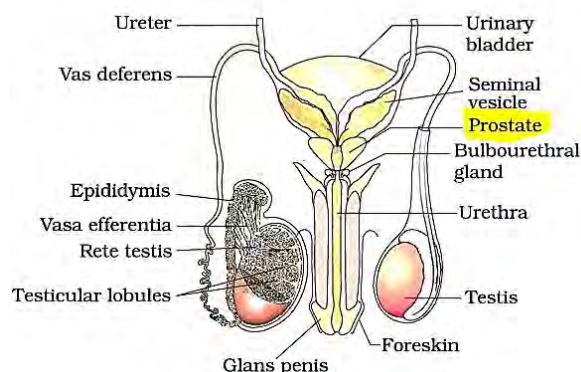
135. (2)

In dicots, the zygote gives rise to the proembryo and subsequently to the globular, heart-shaped and mature embryo.

(NCERT Ed. 2025-26, 12th, Page No. 19)

136. (4)

In the given diagram labelled part X represents prostate gland.



Prostate is situated inferior to the urinary bladder and surround origin of urethra. It secretes prostatic fluid that contain several substances. Prostatic secretion forms about 25% of the volume of semen. It is slightly acidic, pH 6.5 due to the presence of citric acid. Prostatic secretion contains substances important for sperm mobility. The secretions involve calcium ions and proteolytic enzymes, e.g., fibrinolysin.

(NCERT Ed. 2025-26, 12th, Page No. 28)

137. (1)

Leydig cells, also known as interstitial cells, are located in the connective tissue (the interstitium) that lies between the seminiferous tubules, not inside them. Their primary function is to produce androgens, like testosterone.

(NCERT Ed. 2025-26, 12th, Page No. 31)

138. (2)

List-I	List-II
Zona pellucida	Non-cellular glycoprotein layer surrounding the oocyte
Graafian follicle	Mature follicle with large fluid-filled antrum
Oogonia	Diploid stem cells that multiply by mitosis in foetal ovary
Primary oocyte	Oocyte arrested in prophase-I of meiosis

(NCERT Ed. 2025-26, 12th, Page No. 33)

139. (4)

Prolactin is primarily secreted by the anterior pituitary gland of the mother, it is not a primary hormone secreted by the placenta itself.

(NCERT Ed. 2025-26, 12th, Page No. 37)

140. (1)

While oogenesis does initiate during foetal development, the primary oocytes arrest in Prophase I of meiosis. This arrest can last for decades, from the foetal period until that specific oocyte is selected for maturation after puberty. The arrest at Metaphase II occurs after ovulation and just before fertilization.

(NCERT Ed. 2025-26, 12th, Page No. 32)

141. (3)

After implantation, finger-like projections appear on the trophoblast called chorionic villi which are surrounded by the uterine tissue and maternal blood. The chorionic villi and uterine tissue become interdigitated with each other and jointly form a structural and functional unit between developing embryo (foetus) and maternal body called placenta.

(NCERT Ed. 2025-26, 12th, Page No. 37)

142. (1)

- The LH surge is absolutely required for ovulation. Without this sharp peak in LH, the mature Graafian follicle will not rupture to release the oocyte. In the context of a normal menstrual cycle, where a mature, responsive follicle is present, the LH surge is the direct and final trigger that is sufficient to cause ovulation within 24-36 hours. While a mature follicle must be present for it to act upon, the surge itself is the definitive event that initiates the process.
- While FSH is crucial for recruiting follicles and stimulating their initial growth, the later stages of follicular maturation are highly dependent on both FSH and LH. LH is required to stimulate the theca cells of the follicle to produce androgens, which are then converted into estrogen by the granulosa cells (a process stimulated by FSH). This estrogen is vital for the follicle's continued growth.
- The actions of FSH and LH are synergistic and interdependent. They work together in a coordinated manner to ensure proper follicular development and estrogen production.

(NCERT Ed. 2025-26, 12th, Page No. 35)

143. (4)

- The thickening of the endometrium (the proliferative phase) is directly stimulated by estrogen, which is produced by the developing ovarian follicles. While FSH and LH stimulate the follicles to grow and produce estrogen, they do not directly act on the endometrium to cause thickening.
- The follicular phase is named for this very reason. Under the influence of FSH, several follicles begin to grow, but typically one becomes dominant and matures into a Graafian follicle, which is ready for ovulation.
- Around the middle of the cycle (day 14), a rapid spike in the level of Luteinizing Hormone (LH), known as the LH surge, is the direct trigger that causes the mature Graafian follicle to rupture and release the ovum (ovulation).
- The secretory phase (or luteal phase), which follows ovulation, is primarily regulated by progesterone. Progesterone is secreted by the corpus luteum (formed from the ruptured follicle) and prepares the endometrium for a potential pregnancy by making it more vascular and glandular. While some estrogen is also present, progesterone is the dominant hormone.
- The follicular phase begins with menstruation and ends with ovulation. It is the luteal/secretory phase that ends with the onset of menstruation if fertilization and implantation do not occur. The drop in progesterone and estrogen levels from the degenerating corpus luteum triggers menstruation.

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144. (1)

The most significant lab finding is substantially lowered testosterone. Testosterone is produced by the Leydig cells (also called interstitial cells) in the testes. Therefore, the primary damage must be to the Leydig cells.

Luteinizing Hormone (LH) from the pituitary gland stimulates Leydig cells to produce testosterone. Follicle-Stimulating Hormone (FSH) from the pituitary stimulates Sertoli cells to support spermatogenesis. In this case, LH and FSH are normal. This indicates that the pituitary gland is functioning correctly. The problem lies within the testes themselves (a condition known as primary hypogonadism). The low testosterone is due to the Leydig cells failing to respond to the normal LH signal.

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145. (1)

The corpus luteum is essential for the maintenance of pregnancy in its initial weeks in humans. The corpus luteum synthesizes progesterone, which supports vascularization and secretory activity in the endometrium, facilitating implantation and nutrition for the developing embryo. Progesterone from the corpus luteum is critical for sustaining the endometrial lining post-implantation and supporting pregnancy until placental hormone production is established.

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146. (3)

The process of meiosis in females begins with a diploid cell called a primary oocyte. The first meiotic division is a reductional division, meaning the chromosome number is halved from diploid (2n) to haploid (n). A key feature of oogenesis is that the cytoplasm is not divided equally. One of the resulting haploid cells, the secondary oocyte, receives almost all of the cytoplasm and organelles. This is important to provide nutrients for a potential embryo. The other haploid cell, the first polar body, is very small and contains little more than a nucleus. It is generally non-functional and will eventually degenerate.

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147. (1)

Follicle-Stimulating Hormone (FSH) is a gonadotropin released by the anterior pituitary gland. Its primary function in females is to stimulate the growth and maturation of ovarian follicles in the ovary.

In this case the patient has low FSH levels. Because FSH is low, the ovarian follicles are not receiving the necessary signal to grow and mature (non-maturing follicles). Without follicular maturation, ovulation cannot occur, and the normal cyclical production of estrogen and progesterone is disrupted. This hormonal imbalance leads to irregular menstruation and the inability to release an egg for fertilization, causing difficulty in conceiving. The root cause of the entire cascade of symptoms is the insufficient production of FSH. Since FSH is produced by the pituitary gland, the dysfunction must lie there.

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148. (1)

List-I	List-II
Morula	8-16 cell embryo
Blastocyst	Characterised by formation of a cavity known as blastocoel
Trophoblast	Outer layer of blastocyst which gets attached to the endometrium
Inner cell mass	Inner layer of blastocyst that gets differentiated as the embryo

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149. (4)

- Rete testis → It is a network of tubules that collects sperm from the seminiferous tubules. The function described (storage and maturation) belongs to the epididymis.
- Vasa efferentia → These are ducts that connect the rete testis to epididymis.
- Epididymis → It is a single, long, coiled tube for sperm maturation and storage. The description of "short ducts that drain sperm from the testis" refers to the vasa efferentia.
- Vas deferens → It is the muscular tube that carries sperm from the tail of the epididymis up to the ejaculatory duct. During ejaculation, its muscular walls contract to propel the sperm forward.

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150. (4)

The hymen is a thin membrane that may partially cover the vaginal opening. Its presence, absence, or state is not a reliable medical or scientific indicator of virginity. It can be torn or stretched by many non-sexual activities (like sports, using tampons) and some women are born without a prominent hymen.

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151. (2)

Accessory Reproductive Glands: These are the glands that produce the fluid component of semen, which mixes with the sperm. Their secretions nourish, protect, and help transport the spermatozoa. The three main accessory glands in males are:

Seminal vesicles: Produce a thick, alkaline fluid rich in fructose that makes up about 60-70% of the semen volume.

Bulbourethral glands (Cowper's glands): Secrete a clear fluid that lubricates the urethra and neutralizes any residual acidic urine.

Prostate gland: Secretes a milky, slightly acidic fluid containing enzymes that help activate sperm.
Seminiferous Tubules: These are the long, coiled tubules located inside the testes. Their function is spermatogenesis, which is the process of producing sperm. They are the site of sperm formation, not a gland that adds fluid to the semen. Therefore, they are a primary reproductive structure, not an accessory gland.

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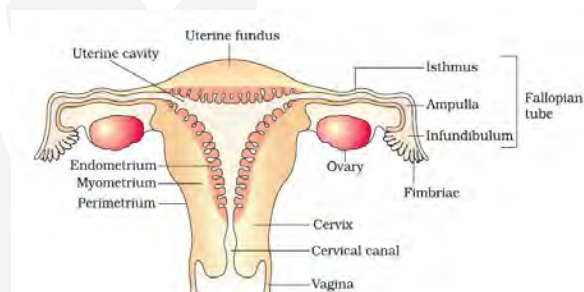
152. (4)

The hypothalamus secretes GnRH, which acts directly on the anterior pituitary gland, causing it to release the gonadotropins, Follicle-Stimulating Hormone (FSH) and Luteinizing Hormone (LH). To maintain hormonal balance, high levels of testosterone in the blood signal back to the brain. Testosterone inhibits the hypothalamus from releasing more GnRH and also directly inhibits the anterior pituitary's sensitivity to GnRH, thus reducing the secretion of LH and FSH. This is a classic negative feedback loop.

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153. (4)

In the given diagram labelled part X represents cervical canal.



The cervical canal forms lower uterine opening, contributing to the birth canal and serving as a gate for sperm entry and menstrual flow exit.

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154. (2)

Colostrum is the first form of milk produced by the mammary glands immediately following delivery of the newborn. Its most critical feature is its high concentration of antibodies, particularly Immunoglobulin A (IgA).

This is vital because a newborn's immune system is immature and not yet fully capable of fighting off infections. The antibodies in colostrum provide passive immunity, coating the baby's digestive tract and protecting it from pathogens in the new environment.

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155. (1)

- The scrotum maintains the testes at a temperature approximately 2.5°C lower than the core body temperature, optimizing conditions for spermatogenesis.
- Spermatogonia are the diploid (2n) stem cells in the testes that serve as the starting point for spermatogenesis. They undergo mitosis to replicate and to produce primary spermatocytes, which then enter meiosis.
- The proliferative phase of the uterine cycle, where the endometrium thickens, is driven by rising levels of estrogen, which is secreted by the developing ovarian follicles. Progesterone levels are low during this phase and only rise after ovulation to initiate the secretory phase.
- The menstrual cycle is governed by the Hypothalamic-Pituitary-Ovarian (HPO) axis. The hypothalamus releases GnRH, the pituitary releases FSH and LH, and the ovaries release estrogen and progesterone. These hormones act on the endometrium and also provide feedback to the hypothalamus and pituitary, creating a tightly regulated cyclical system.

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156. (4)

- For fertilization to happen, both the sperm and the ovum must be viable and present in the ampullary region of the fallopian tube at the same time. This is why coitus during the ovulatory phase of the cycle has the highest chance of leading to pregnancy.
- After one sperm successfully fuses with the ovum's plasma membrane, it triggers a series of events (the cortical reaction) that hardens the zona pellucida. This "block to polyspermy" prevents any other sperm from entering, ensuring the correct diploid chromosome number is restored.
- The head of the sperm contains a cap-like structure called the acrosome, which is filled with hydrolytic enzymes. These enzymes are released to help the sperm digest its way through the protective outer layers of the ovum (the corona radiata and the zona pellucida), allowing it to reach and fuse with the egg's membrane.
- Fertilization, the fusion of the sperm and ovum, takes place in the ampullary region of the fallopian tube (oviduct), not the uterus. The resulting zygote then travels down the fallopian tube for several days, undergoing cell division, before implanting in the wall of the uterus at the blastocyst stage.

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157. (1)

- In a non-pregnant cycle, the corpus luteum degenerates after about 10-14 days, leading to a drop in progesterone and the start of menstruation. However, if an embryo implants in the uterine wall, its outer layer (the syncytiotrophoblast, a precursor to the placenta) begins to produce hCG. This hormone mimics the action of LH, signaling the corpus luteum to continue producing progesterone. This continued progesterone production is vital for maintaining the endometrial lining and sustaining the early pregnancy until the placenta can produce enough progesterone on its own.
- Because hCG is produced by the developing embryo and enters the mother's bloodstream, its levels rise rapidly after implantation. The hormone is filtered by the kidneys and excreted in the urine. Therefore, its presence in either blood or urine is a definitive marker of pregnancy. Both laboratory blood tests and over-the-counter urine tests are designed to detect the presence of hCG.

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158. (2)

List-I	List-II
Spermiogenesis	Maturation of spermatids
Spermiation	Release of sperms into lumen of seminiferous tubule
Oogenesis	Formation of female gamete
Menarche	First menstruation

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159. (3)

- The head of a human sperm contains a highly compacted nucleus with a haploid set of chromosomes. This means it has 23 chromosomes, which, upon fertilization, will combine with the 23 chromosomes from the female egg to form a diploid zygote with 46 chromosomes.
- While the tail, or flagellum, is responsible for the sperm's motility through its whip-like, corkscrew motion, the energy for this movement is generated by the mitochondria located in the middle piece, not the tail itself.
- The plasma membrane is a continuous outer boundary that envelops the entire sperm cell, including the head, middle piece, and tail. This membrane plays a critical role in protecting the sperm and mediating interactions with its environment and the egg.

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160. (3)

List-I (Reproductive Event)	List-II (location/feature)
Spermatogenesis	Seminiferous tubule
Oogenesis	Ovary
Fertilisation	Ampullary region of fallopian tube
Implantation	Uterus

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161. (1)

Maintenance of hygiene and sanitation during menstruation is very important. Take bath and clean yourself regularly. Use sanitary napkins or clean homemade pads.

Change sanitary napkins or homemade pads after every 4–5 hrs as per the requirement as it is very important to prevent infections. Dispose of the used sanitary napkins properly wrapping it with a used paper

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162. (4)

Luteinizing Hormone (LH) is a gonadotropin whose primary roles in the female reproductive cycle are to trigger ovulation (the LH surge) and to maintain the corpus luteum. During late pregnancy and labor, LH levels are very low and it plays no direct or required role in initiating or sustaining uterine contractions. The key hormones for labor are oxytocin and prostaglandins.

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163. (1)

List-I (Hormone)	List-II (Function)
Estrogen	Regulates menstrual cycle and secondary sexual characters
Progesterone	Maintains endometrium and supports pregnancy
Follicle stimulating hormone	Stimulates growth and maturation of ovarian follicles
Oxytocin	Causes uterine contractions during childbirth

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164. (2)

The female external genitalia include mons pubis, labia majora, labia minora, hymen and clitoris. Mons pubis is a cushion of fatty tissue covered by skin and pubic hair. The labia majora are fleshy folds of tissue, which extend down from the mons pubis and surround the vaginal opening. The labia

minora are paired folds of tissue under the labia majora. The opening of the vagina is often covered partially by a membrane called hymen. The clitoris is a tiny finger-like structure which lies at the upper junction of the two labia minora above the urethral opening.

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165. (3)

- Follicle-Stimulating Hormone (FSH) targets the Sertoli cells within the seminiferous tubules, stimulating them to secrete substances that are essential for the process of spermiogenesis (the maturation of spermatids into spermatozoa).
- Luteinizing Hormone (LH) acts on the Leydig (or interstitial) cells, which are located between the seminiferous tubules, stimulating them to produce and secrete androgens, primarily testosterone.
- Androgens, with testosterone being the most important, are crucial for stimulating and maintaining the entire process of spermatogenesis.
- Testosterone is the primary androgen, which is a type of steroid hormone. Follicle-Stimulating Hormone (FSH), however, is a gonadotropin, which is a peptide hormone, not an androgen.
- Spermatogenesis is the entire process of sperm production. Spermiation is the final step of this process, where the mature spermatozoa are released from the Sertoli cells into the lumen of the seminiferous tubule. Therefore, it is a part of spermatogenesis.

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166. (4)

List-I	List-II
Fimbriae	Help collect the ovum after ovulation
Infundibulum	Funnel-shaped opening of oviduct near the ovary
Sertoli cells	Present inside the seminiferous tubule
Interstitial cells	Secrete testosterone

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167. (3)

The uterus is single and it is also called womb. The shape of the uterus is like an inverted pear. It is supported by ligaments (not tendons) attached to the pelvic wall. The uterus opens into vagina through a narrow cervix. The cavity of the cervix is called cervical canal which alongwith vagina forms the birth canal.

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168. (2)

The developmental journey from a single cell to an implanted embryo follows a precise and unchangeable order. The process begins with the zygote, the initial product of fertilization. Through rapid cell division, this single cell compacts into a solid morula. A crucial reorganization then occurs as a fluid-filled cavity forms, transforming the solid morula into the more complex blastocyst, the stage required for implantation. Only after implantation does the inner cell mass of the blastocyst begin to differentiate into the recognizable embryo. This logical progression from a simple single cell to a complex, structured organism confirms that the sequence Zygote → Morula → Blastocyst → Embryo is the only correct developmental pathway.

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169. (2)

The duration of human pregnancy is calculated from the first day of the mother's last menstrual period (LMP). However, a normal, healthy delivery can occur over a range of time. The period from 36 weeks to 40 weeks is generally considered term.

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170. (4)

- Fertilization is the process of male and female gamete fusion to form a diploid zygote.
- After a sperm successfully penetrates the zona pellucida and fuses with the oocyte's plasma membrane, it triggers a reaction (the cortical reaction) that causes the zona pellucida to harden. This change prevents other sperm from entering, a crucial mechanism known as the block to polyspermy.
- The zygote undergoes several rounds of cell division (cleavage) to form a solid ball of cells called the morula. The morula then develops into a blastocyst, which is a hollow ball of cells with an inner cell mass. The correct sequence is Zygote → Morula → Blastocyst.
- The uterine wall has three layers: the perimetrium (outermost), the myometrium (middle, muscular layer), and the endometrium (innermost lining). Implantation is the process where the blastocyst burrows into and attaches to the endometrium, the innermost layer, not the outermost one.
- Sex is determined at the moment of fertilization. The female gamete (ovum) always carries an X chromosome. The male gametes (sperm) can carry either an X or a Y chromosome. If an X-carrying sperm fertilizes

the egg, the resulting zygote is female (XX). If a Y-carrying sperm fertilizes the egg, the zygote is male (XY). Therefore, the type of sperm that fertilizes the egg determines the sex, which happens at fertilization, not during the formation of the gametes themselves.

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171. (2)

- The penis is the male external genitalia. It is made up of special tissue that helps in erection of the penis to facilitate insemination.
- The male accessory glands include paired seminal vesicles, a prostate and paired bulbourethral glands. Secretions of these glands constitute the seminal plasma which is rich in fructose, calcium and certain enzymes.

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172. (4)

- The placenta is a major endocrine organ that produces several crucial hormones. It produces human Chorionic Gonadotropin (hCG), human Placental Lactogen (hPL), and steroid hormones like estrogens and progesterone. However, oxytocin is produced by maternal pituitary gland during parturition.
- The placenta is a unique organ because it is composed of tissues from two different individuals. It has a foetal portion (the chorion frondosum) that develops from the blastocyst, and a maternal portion (the decidua basalis) that develops from the mother's uterine lining (endometrium).
- The placenta facilitates exchange, but maternal and foetal blood do not mix directly. They are separated by the placental barrier (membrane). The chorionic villi are bathed in maternal blood, but the blood supplies remain distinct to prevent immune reactions and other complications.

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173. (4)

The human male ejaculates about 200 to 300 million sperms during a coitus of which, for normal fertility, at least 60 per cent sperms must have normal shape and size and at least 40 per cent of them must show vigorous motility.

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174. (2)

The endometrium is the innermost mucosal layer of the uterine wall. Its most unique and defining feature is its dynamic nature, as it constantly changes in response to the hormonal fluctuations of the menstrual cycle. Therefore, a high count alone is not sufficient.

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175. (3)

Secretions of epididymis, vas deferens, seminal vesicle and prostate are essential for maturation and motility of sperms. The seminal plasma along with the sperms constitute the semen. The seminiferous tubules are the site of sperm production (spermatogenesis), not a primary contributor of secretions for seminal plasma or sperm maturation.

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176. (3)

The wall of the uterus has three layers of tissue. The external thin membranous perimetrium, middle thick layer of smooth muscle, myometrium and inner glandular layer called endometrium that lines the uterine cavity. The endometrium undergoes cyclical changes during menstrual cycle while the myometrium exhibits strong contraction during delivery of the baby.

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177. (1)

List-I	List-II
Corpus luteum	Secretes progesterone
Primary follicle	Single layer of granulosa cells around primary oocyte
Secondary follicle	Multiple layers of granulosa cells; formation of theca begins
Tertiary follicle	Presence of fluid-filled antrum; well-developed theca layers

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178. (2)

The mammary glands are paired structures (breasts) that contain glandular tissue and variable amount of fat. The glandular tissue of each breast is divided into 15-20 mammary lobes containing clusters of

cells called alveoli. The cells of alveoli secrete milk, which is stored in the cavities (lumens) of alveoli. The alveoli open into mammary tubules. The tubules of each lobe join to form a mammary duct. Several mammary ducts join to form a wider mammary ampulla which is connected to lactiferous duct and opens at the nipple, from where milk is expressed during breastfeeding.

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179. (1)

In human beings, after one month of pregnancy, the embryo's heart is formed. The first sign of growing foetus may be noticed by listening to the heart sound carefully through the stethoscope. By the end of the second month of pregnancy, the foetus develops limbs and digits. By the end of 12 weeks (first trimester), most of the major organ systems are formed, for example, the limbs and external genital organs are well developed. The first movements of the foetus and appearance of hair on the head are usually observed during the fifth month. By the end of about 24 weeks (end of second trimester), the body is covered with fine hair, eye-lids separate, and eyelashes are formed. By the end of nine months of pregnancy, the foetus is fully developed and is ready for delivery.

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180. (3)

- There are remarkable differences between the reproductive events in the male and in the female.
- The sperm formation continues even in old men, but formation of ovum ceases in women around the age of fifty years not sixty years.

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